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Marked Engaged, Opens Anyway: Validating a Household Safety Walkthrough's Latch-Engaged Checklist Item Against a Standardised Pull-Force Test

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Abstract. Home safety walkthroughs — conducted by parenting networks, paid childproofing installers, or a household's own checklist — typically ask whether a kitchen cabinet child-safety latch is engaged and functional, and record the answer as an adequacy signal rather than as a measurement of the latch itself. We report a validation study of 412 latches across 96 households, comparing each latch's walkthrough-recorded status against a standardised pull-force test administered within one week of the walkthrough. The checklist records 93.9% of latches as engaged and functional; the pull-force test passes only 64.6%. Among latches the checklist marked engaged, 32.3% fail the pull-force test outright, a disagreement a paired test finds far from chance (McNemar's $\chi^2(1, N = 412) = 113.5, p < .001$). Failure concentrates in adhesive-mounted latches, whose bond has usually loosened rather than the mechanism itself having failed. An ablation comparing latches originally checked by a paid professional installer against those self-checked by a household member finds no reliable difference in subsequent pull-force performance (Mann- Whitney $U = 1481$, $p = .58$). We conclude the checklist item functions less as a measurement of the latch than as a record that a latch was once clipped into place.

Introduction

Household cleaning products stored under a kitchen sink remain a leading and persistent source of emergency-department visits among young children [1], and the kitchen cabinet child-safety latch is the device most households install specifically to keep that cabinet closed. It is among the most common safety devices installed in a home with young children, and among the least frequently re-examined once installed. It appears on nearly every home safety walkthrough in some form — a self-administered checklist, a paid childproofing installer's sign-off, a parenting network's peer visit — almost always as a single binary item: engaged and functional, yes or no. Validation research on home safety questionnaires generally finds that agreement between a respondent's reported practice and what an independent observer finds ranges widely across items, from substantial to barely better than chance [2], [3]. Checklist instruments built specifically for the home environment have been validated for content — whether their items are relevant and clearly worded — without validating whether a positive item response corresponds to the hazard the item is meant to control [4], [5]. Adjacent domains report the same gap in sharper relief: a standardised checklist administered by a certified technician still leaves the large majority of child car-seat installations short of fully correct [6], and a surgical safety checklist's documented completion does not guarantee the step it records was actually performed as intended [7], [8].

Whether the same gap holds for the humblest safety device in the house — a latch that either physically resists a pull or does not — has not, to our knowledge, been tested directly against the latch's own mechanical performance. We treat the cabinet latch as a minimal case of a broader pattern in household and institutional self-assessment: a checklist item recording that a device was once installed, standing in for a measurement of whether the device still functions.

Our contributions:

- We report, to our knowledge, the first study to validate a home safety walkthrough's latch-engaged checklist item directly against a standardised mechanical pull-force test on the same latch.
- We show the checklist substantially over-credits functional status relative to the pull-force test, with the disagreement concentrated in a single mechanism type and a small number of recurring failure modes.
- We find no reliable difference in subsequent latch performance between professionally installed and self-checked latches, suggesting the gap is a property of the checklist item rather than of who is completing it.

The remainder of the paper reviews related work on checklist validity across domains (Related work), describes our walkthrough-matched pull-force protocol (Method), reports the checklist/pull-force disagreement and its failure composition (Results), and discusses what a latch-engaged item appears to record instead of measuring (Discussion).

Related work

Home safety questionnaire validation work finds agreement between self-reported and independently observed practice varies sharply by item, with some practices — notably fixed installations like stair gates — showing substantial agreement and others showing much weaker correspondence [2], [3]. Checklist development for the home environment has mostly pursued content validity, establishing that an item is relevant and its wording clear to a panel of experts, which is a distinct property from criterion validity against the hazard the item is meant to track [4], [5]. The same distinction recurs in institutional safety auditing: audit corrective actions are more often symbolic than tied to the actual source of harm they were meant to address [9], and a systematic review of surgical safety checklist use finds implementation completeness holds at only around half of the checklist's own items even where overall compliance is reported as high [8], consistent with earlier qualitative work finding a checklist's documented application diverges from what direct observation shows occurred [7].

Device-specific literature suggests the same pattern where a device, not a process, is being certified. A national sample of child car-seat installations assessed by certified technicians against a standardised checklist finds barely one in eight installations entirely correct despite the technician's own sign-off on individual components [6]. Consumer safety packaging research finds self-reported closure compliance for a child-resistant pack and sensor-measured closure compliance for the same pack diverge substantially [10], and the broader child-resistant packaging literature — while establishing that a correctly functioning closure measurably reduces unintentional poisoning — has long noted that effectiveness depends on the closure being engaged correctly rather than merely present [11], [12]. Prior work at this institution validated a self-report pace scale against stopwatch-timed laps and found the two agreed only moderately [13], and found shoppers' stated trust in a self-checkout's error prompt did not predict their actual override behaviour [14]; the present paper extends that pattern from a leisure self-report instrument to a safety-relevant mechanical fixture. Peltzman's account of behavioural offset following automobile safety regulation is a standing caution that a household treating a latch as handled once it is checked "yes" may adjust its own vigilance accordingly [15]. Our pull-force protocol draws on instrumented pulling-task and grip-strength measurement developed for infants and toddlers to calibrate a plausible test threshold [16], [17].

Method

We recruited 96 households with a home safety walkthrough completed within the preceding three months, drawing on parenting networks, a community child-safety program, and paid childproofing installation services, and restricted the sample to walkthroughs whose checklist included an "engaged and functional" item for kitchen cabinet or drawer child-safety latches. This yielded 412 latches across four mechanism types in common use: adhesive strap, magnetic (concealed), spring-clip, and sliding U-latch. For each latch we recorded the walkthrough's checklist status (engaged and functional: yes or no) and, where recorded, whether the original installation was checked by a paid professional installer or by a household member.

Within one week of the walkthrough, a research assistant blind to the checklist status administered a standardised pull-force test to each latch: a calibrated spring-scale device applied a steadily increasing pull along the latch's normal release axis, recording the force at which the latch released or reached a test ceiling. The pass threshold was set at 30 N, informed by instrumented pulling-task protocols developed to measure maximal voluntary pulling force in infants and toddlers aged 6-36 months [16] and by population norms for children's manual grip strength across early childhood [17], placing the threshold near the low end of forces a strong-pulling toddler in this age range can plausibly generate.

Agreement between checklist status and pull-force outcome, both recorded per latch, is assessed with McNemar's test for paired binary classifications:

$$\chi^2 = \frac{(b-c)^2}{b+c}$$

where b is the count of latches marked engaged that fail the pull-force test and c is the count marked not engaged that pass it — the two disagreement cells, holding constant the cells where checklist and test agree.

Latches that failed the pull-force test were independently coded by two research assistants into five mutually exclusive failure-mode categories (Table 1), developed inductively from a pilot round of twenty latches and then applied to the full set of failures; a second coder independently coded a 30% random subsample, yielding acceptable agreement (Cohen's $\kappa = 0.79$).

Failure mode	Operational definition
Bond loosened	An adhesive-mounted latch's strap or plate separates from its mounting surface under load before the mechanism itself disengages.
Tension fatigued	A spring or clip mechanism releases below threshold with its mounting intact, consistent with reduced return tension.
Misaligned	The latch's two halves no longer meet squarely, traced to a documented cabinet re-hang, repaint, or hardware replacement since installation.
Never closed correctly	The latch was not seated in its fully closed position at the time of testing, with no evidence of subsequent wear or tampering.
Bypassed	The mechanism shows evidence of deliberate removal or defeat — a strap cut, a magnet relocated — reported by a household member as an adult's own workaround.

Table 1: Coding rubric for pull-force failure modes; each failing latch received exactly one category.

For the 412 latches where the original checker was recorded, we compare the continuous pull-force release point between latches checked by a paid professional installer ($n = 176$) and those self-checked by a household member ($n = 236$) with a Mann-Whitney U test, treating the split as a naturally occurring ablation on whether professional verification changes the correspondence between recorded and actual latch function.

Results

Across the 412 latches, the walkthrough checklist recorded 387 (93.9%) as engaged and functional and 25 (6.1%) as not. The pull-force test told a different story: only 266 latches (64.6%) reached the 30 N threshold before releasing. Figure 1 breaks this gap down by mechanism type: the checklist rate holds above 90% for every type, while the pull-force pass rate ranges from 47.8% for adhesive-strap latches to 79.8% for sliding U-latches — the only type where checklist and test come within twenty points of each other.

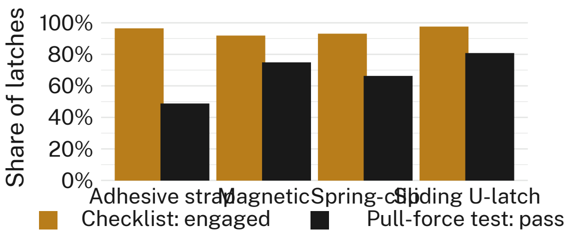

Figure 1: Checklist "engaged and functional" rate versus pull-force test pass rate, by latch mechanism type. Adhesive-strap latches carry the widest gap, at 47.7 points.

Treated as a paired classification, the disagreement is concentrated almost entirely in one direction. Of the 387 latches the checklist marked engaged, 262 (67.7%) pass the pull-force test and 125 (32.3%) fail it outright. Of the 25 latches the checklist marked not engaged, 21 (84.0%) fail as expected and 4 (16.0%) pass. McNemar's test on the two disagreement cells is decisively non-null ($\chi^2(1, N = 412) = 113.5, p < .001$): a latch the checklist calls engaged is roughly thirty times more likely to be a false positive than a latch it calls unengaged is to be a false negative.

Among the 146 failing latches, failure composition is dominated by a single mode (Figure 2): 37.7% failed because an adhesive mount had loosened or peeled away from the cabinet surface, ahead of tension fatigue in spring or clip mechanisms (24.0%), misalignment traced to a documented re-hang, repaint, or hardware swap (19.2%), a latch never closed correctly at installation (12.3%), and a mechanism an adult household member had deliberately bypassed (6.8%).

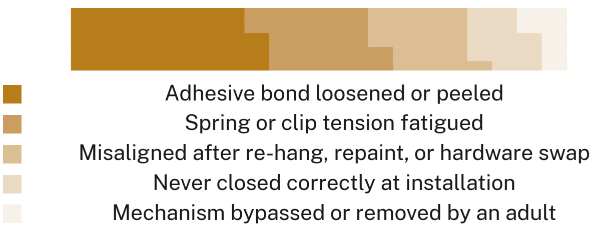

Figure 2: Composition of failure modes among 146 failing latches. A loosened adhesive bond accounts for more failures than the next two categories combined.

The ablation comparing who originally checked the latch finds no reliable difference in subsequent pull-force performance. Professionally checked latches release at a median of 27 N against 25 N for self-checked latches; the full distributions, shown in Figure 3, overlap almost completely (Mann-Whitney $U = 1481$, $p = .58$). Pass-rate at the 30 N threshold tells the same story at a coarser grain: 66.5% for professionally checked latches versus 63.1% for self-checked latches, a difference a chi-square test does not distinguish from chance ($\chi^2(1, N = 412) = 0.48$, $p = .49$).

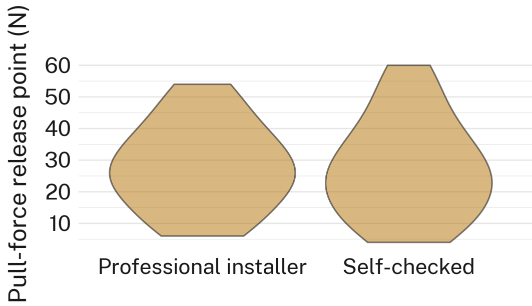

Figure 3: Pull-force release point by original checker. Professional installation shifts the median release force by under 2 N.

Discussion

The clearest finding here is not that cabinet latches fail — mechanisms wear, adhesives loosen, cabinets get repainted — but that the checklist item built to catch this holds almost no diagnostic relationship to it. A latch marked engaged is nearly as likely to fail a direct pull-force test as a coin flip weighted toward passing, and the item's rare "not engaged" mark is reserved for latches obviously never closed at all rather than for latches whose function has since degraded. This reads as consistent with the pattern Hutchinson and colleagues describe in institutional safety auditing, where a recorded corrective action is more often a record that the audit occurred than a treatment of the hazard it names [9], and with checklist research in surgical settings, where documented completion and directly observed application are shown to diverge [7], [8]. The ablation's null result is informative in the same direction: if the gap were mainly a matter of who checks the box, professional installation should have narrowed it, and it does not. The checklist item appears to certify an installation event rather than a current state, which is consistent with our own institution's prior finding that a self-report instrument and a directly measured one can drift apart without anyone along the way noticing [13], [14]. Whether a household that reads "engaged and functional: yes" then relaxes the vigilance the latch was meant to substitute for — a Peltzman-style offset — is a plausible extension of these results that our design cannot test directly [15].

Limitations

Our sample is drawn from households reachable through parenting networks and childproofing services and that agreed to a research pull-force test on their own safety hardware, so households for whom a latch's failure would be most consequential may be over-or under-represented in ways we cannot fully rule out. The 30 N pass threshold is a single cut point informed by pulling-task and grip-strength literature developed for a related but not identical task, and a different threshold would shift the absolute pass rate without disturbing the checklist-versus-test disagreement the paired test detects. Failure-mode coding relied on visual and reported evidence rather than material testing of the adhesive or spring components themselves, though inter-coder agreement on the resulting categories was acceptable.

Conclusion

A household safety walkthrough's "engaged and functional" latch item shows almost no correspondence with whether the latch it describes actually resists a standardised pull, over-crediting function most severely for the mechanism type — adhesive-mounted latches — that the market currently sells most often as the low-cost default. Neither professional installation nor self-checking narrows the gap, suggesting the item measures an installation event rather than the ongoing state that matters. Future work might test whether re-phrasing the checklist item around a household's own last direct test of the latch, rather than its installed presence, narrows the disagreement, and whether the same pattern holds for other single-item device checks — a smoke alarm's test button, a fire extinguisher's gauge — that a walkthrough marks in passing rather than measures directly.

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